LivreAnálise: A Free and Open Source Desktop Tool for Qualitative Data Analysis

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Abstract. *Abstract provisório.*

1. Context

Qualitative coding is a central activity in Information Systems research. The researcher reads transcripts line by line, assigns codes to excerpts, groups codes into categories and themes, and revises that structure several times before arriving at a final set of findings [Braun and Clarke 2006]. Doing this by hand, in a word processor or a spreadsheet, does not scale past a handful of interviews.

Software support for this activity exists, but it is almost entirely commercial. The established tools in the field, such as ATLAS.ti, NVivo and MAXQDA, are proprietary and sold under paid licenses. This puts them out of reach of a large part of the Brazilian academic community, and particularly of undergraduate and early graduate students, who need the tool for a single piece of work and have no funding line to cover it.

The first author faced exactly this problem while writing his undergraduate thesis. The analysis was carried out in ATLAS.ti, a paid tool, not by preference but by elimination: the free alternatives available at the time offered few features and dated interfaces.

Beyond cost, two further constraints shaped the problem. First, most contemporary alternatives are cloud services, which requires uploading research material to third-party infrastructure. When that material is an interview with an identifiable participant, this is not a minor operational detail. Second, none of the tools we examined offered a way to use general-purpose AI assistants alongside the analysis without handing the entire dataset over to a model provider.

This paper reports the development of LivreAnálise, a free and open source desktop application for qualitative coding that runs entirely on the researcher's own machine. Section 2 describes the process and the architecture adopted, and Section 3 presents the resulting tool and how it has been used so far.

2. Adopted Process

Development started in June 2026 and has followed an iterative and incremental approach, with four public releases to date; the version described here is 0.7.0, from October 2026. The team is small: two developers, one of whom is the first author and acts as maintainer, and one supervising researcher. External contributions are made through pull requests

against the public repository, and roughly a fifth of the commits so far come from a contributor other than the maintainer.

Three design decisions were taken at the outset and have constrained every choice since.

The application runs locally, with no account and no server. A project is a single file on the researcher's disk, so there is no infrastructure to pay for and no data leaving the machine. This also means the tool works offline.

The application embeds no AI model. Rather than integrating a provider, which would imply a cost and a data transfer the researcher may not want, the tool exports the analysis in a form that any external assistant can read. The integration is manual and deliberate: the researcher decides what leaves the machine, and when.

The project format is interoperable. Adopting the open REFI-QDA exchange format [REFI-QDA 2019] means a project can be moved in and out of ATLAS.ti, so choosing the tool is not a commitment that locks the researcher in.

The implementation is an Electron application written in TypeScript, with React and Tailwind in the interface. Project data is stored in SQLite, accessed through better-sqlite3 and Drizzle, in a single `.liva` file. Transcription uses whisper.cpp with quantized Whisper models [Radford et al. 2023] in ggml format, and a bundled ffmpeg build extracts the audio track. Figure 1 summarizes the architecture.

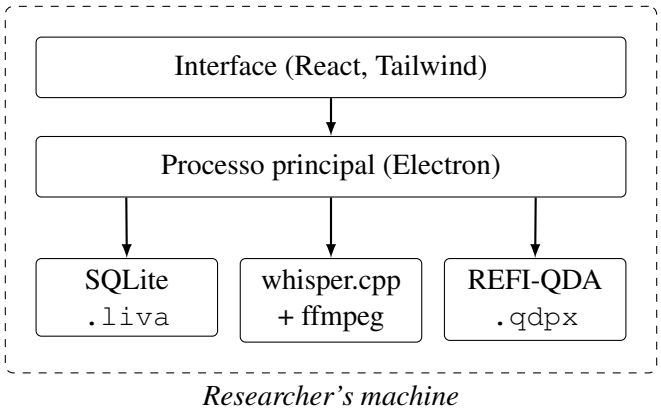


Figure 1. Architecture of LivreAnálise. Every component runs locally.

Each merge into the main branch runs the test suite and the type checker and publishes signed-off installers for Linux, Windows and macOS, so a release is always reproducible from the repository.

The tool has been evaluated in two ways. It is in use in two ongoing undergraduate theses, by the second and third authors, who adopted it from the first public release onwards and whose reports informed successive iterations. A hands-on workshop with a research group outside the development team is also planned, with an evaluation questionnaire at the end.

3. Solution

LivreAnálise supports the full coding cycle in a single desktop application. Documents in `.docx` and plain text are imported into the project, and their content is copied into the project file, so moving or deleting the original afterwards does not break the analysis. The researcher selects an excerpt and applies a code; coded regions are highlighted in the code's colour and labelled in the right margin, aligned to where each excerpt begins, as shown in Figure 2. Overlapping codings of the same code are merged automatically, and the extent of a quotation can be adjusted by dragging its ends. Documents can be edited inside the application without displacing the codings already made.

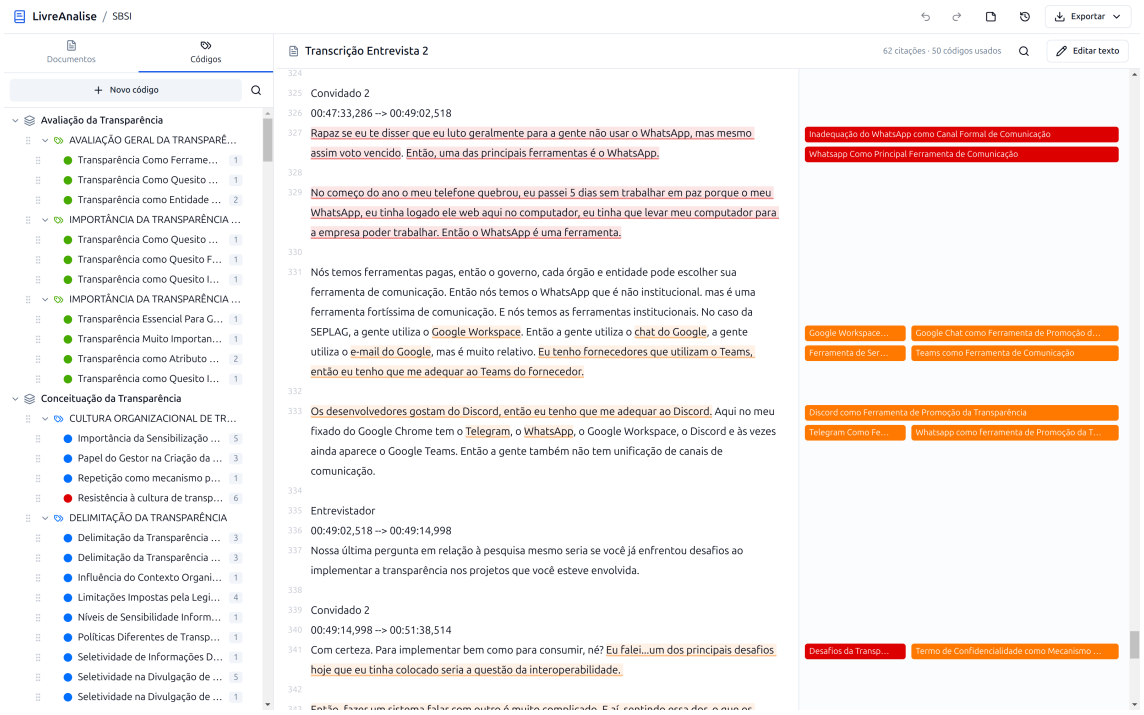


Figure 2. Coding a document: the hierarchy of groups, categories and codes on the left, and the labels aligned on the right margin.

The code hierarchy has exactly three levels. A *code* is the atom and the only level a quotation attaches to. A *category* is a code that contains other codes, a property derived from having children rather than a separate kind of entity, and it cannot be applied to an excerpt. A *group* gathers categories and codes across the hierarchy. Categories and groups are always created from the items they gather, so a category never exists empty. The tool also provides per-code usage counts, a list of every excerpt of a given code, research notes anchored in the text, full undo and redo, named project checkpoints, search across the project, and typographic settings including the OpenDyslexic typeface.

Two features distinguish the tool from the free alternatives we found. Transcription of audio and video runs **locally**, through whisper.cpp, with the text appearing on screen as it is recognized and a progress bar driven by the fraction of audio already processed; nothing is sent to any API. And the **AI export** produces a text file describing the analysis at one of three scopes, from the code structure alone to the full set of documents with their markings, which the researcher can paste into any assistant. The file opens with

an explicit warning that the material will leave the machine and that participant consent and ethics approval should be checked first.

Interoperability with ATLAS.ti is provided through REFI-QDA [REFI-QDA 2019]: a `.qdpx` project can be opened, creating a new project, and any project can be exported back. Codes map to leaf `Code` elements, categories to nested `Code` elements, groups to `Set` and `MemberCode`, and quotations to `PlainTextSelection` and `Coding`.

Anonymization is deliberately out of scope. It is the researcher's responsibility, and the tool expects the material it receives to already be in the form the research ethics process requires.

The benefits of the approach are threefold. There is **no cost**, which removes the barrier that motivated the work. There is **data sovereignty**, since no component requires an account, a server or a network connection, and the material under analysis never leaves the researcher's machine unless the researcher explicitly exports it. And there is **no lock-in**, because the project format is open and the code is public and open to contribution. The tool is available at `https://github.com/Lucas-Tito/LivreAnalise`.

Acknowledgements

References

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